

EXPERIMENT 15

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Aim

To prepare an algorithm and test-case table, and to write, compile, execute, and test a menu-driven C program using **switch-case**. The program accepts a positive integer and checks either whether all its digits are even or whether all its digits are prime.

Problem Statement

Input a positive integer and display a menu with two operations: (1) check whether all digits are even, and (2) check whether all digits are prime. Prime digits are 2, 3, 5, and 7.

Algorithm

1. Start.
2. Input a positive integer n .
3. Display the menu: 1 - All digits even, 2 - All digits prime.
4. Input the user's choice.
5. Use a **switch** statement on the choice.
6. For choice 1, copy n to a temporary variable. Repeatedly obtain the last digit using $digit = temp \% 10$. If any digit is odd, mark the result false; otherwise continue with $temp = temp / 10$.
7. For choice 2, similarly inspect every digit. If a digit is not 2, 3, 5, or 7, mark the result false.
8. Display the appropriate result.
9. For any other menu choice, display 'Invalid choice'.
10. Stop.

Test Case Table

Test	Number	Choice	Expected Output
1	12345	1	All digits are not even
2	24680	1	All digits are even
3	2357	2	All digits are prime
4	2379	2	All digits are not prime
5	2222	1	All digits are even
6	2222	2	All digits are prime
7	4682	2	All digits are not prime
8	2468	3	Invalid choice

C Program

```
#include <stdio.h>

int main()
```

```

{
    int n, temp, digit, choice, flag;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\n1. Check whether all digits are even");
    printf("\n2. Check whether all digits are prime");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            temp = n;
            flag = 1;

            while (temp > 0)
            {
                digit = temp % 10;
                if (digit % 2 != 0)
                {
                    flag = 0;
                    break;
                }
                temp = temp / 10;
            }

            if (flag)
                printf("All digits are even\n");
            else
                printf("All digits are not even\n");
            break;

        case 2:
            temp = n;
            flag = 1;

            while (temp > 0)
            {
                digit = temp % 10;
                if (!(digit == 2 || digit == 3 ||
                    digit == 5 || digit == 7))
                {
                    flag = 0;
                    break;
                }
                temp = temp / 10;
            }

            if (flag)
                printf("All digits are prime\n");
            else
                printf("All digits are not prime\n");
            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}

```

Compilation and Execution

Compile: gcc experiment15.c -o experiment15

Run: ./experiment15

Example run for number 24680 and choice 1:

```
Enter a positive integer: 24680
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are even
```

Example run for number 2357 and choice 2:

```
Enter a positive integer: 2357
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are prime
```

Result

The menu-driven C program was prepared using switch-case, compiled, executed, and tested. It correctly determines whether all digits of a positive integer are even or whether all digits are prime for the prepared test cases.